Botao Zhang

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Education

Fudan University — *Shanghai, China*
B.S. in Mathematics, School of Mathematical Sciences — Sept 2023 – Jul 2027 (expected)
GPA: 3.66

University of California, San Diego — *San Diego, CA, USA*
Exchange Student in Mathematics — Jan 2026 – Mar 2026
GPA: 4.00

Shanghai High School — *Shanghai, China*
Sept 2020 – Jun 2023

Research Experience

Benna Lab, University of California, San Diego — *San Diego, CA, USA*
Visiting Student · Advisor: Prof. Marcus Benna — Jan 2026 – Present

- Developed a biologically plausible neural network that decomposes patterns into features, inspired by SNMF algorithm's derivation from Cengiz Pehlevan's lab. This decomposition allows patterns to be represented sparsely as combinations of features, enabling efficient storage in a classical Hopfield network.

Laboratory of Cognitive Computational Neuroscience and Neuroimaging
Shanghai Jiao Tong University — *Shanghai, China*
Visiting Student · Advisor: Prof. Ruyuan Zhang — Oct 2023 – Dec 2025

- Investigated arbitration between model-free and model-based control in decision-making. Building on the reliability-based arbitration framework of Lee et al. (2014), developed and tested a simpler, equally intuitive arbitration model to explain behavioral data.

Research Interests

(value-based) decision making, planning, reinforcement learning, cognitive maps, problem solving, memory, episodic memory, working memory, metacognition, meta-learning, generalization, representation learning, structure learning, neural dynamics, neural geometry, neural representation, mixed selectivity, neural network modeling, few-shot learning, continual learning, emotion, computational psychiatry, intelligence, complexity science, predictive coding.

Technical Skills

Programming: Python, R, LaTeX
Languages: Mandarin Chinese (native), English (fluent)

Awards

Outstanding Student Scholarship (Third Prize), Fudan University — 2023
Undergraduate Scholarship in Basic Disciplines, Fudan University — 2023, 2024
First Prize, Chinese National High School Mathematics Olympiad, Shanghai — 2021, 2022